DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II

WEEK 1: LECTURE NOTE

LIMITS AND THEIR PROPERTIES

A Preview of Calculus

What is calculus?
Key Takeaways:

- Calculus is a branch of mathematics that involves the study of rates of change.
- All mathematics was static before the invention of calculus; that is, it could only help calculate objects that were perfectly still.
- No objects - from the stars in space to subatomic particles or cells in the body - are always at rest. The universe is constantly moving and changing.
- Invention of calculus helped to determine how particles, stars, and matter actually move and change in real time.
- Calculus is the mathematics of **velocities**, **accelerations**, **tangent lines**, **slopes**, **areas**, **volumes**, **arc lengths**, **centroids**, **curvatures**, and a variety of other concepts that have enabled scientists, engineers, and economists to model real-life situations.
- **Gottfried Leibniz** and **Isaac Newton**, 17th-century mathematicians, both invented calculus independently. Newton invented it first, but Leibniz created the notations that mathematicians use today.
- Differential calculus determines the rate of change of a quantity.
- Integral calculus finds the quantity where the rate of change is known.
- **Practical Applications.** Calculus is used in
  (a) **PHYSICS** in the study of concepts of motion, electricity, heat, light, harmonics, acoustics, and astronomy.
  (b) **Geography**, **computer vision**, **photography**, **artificial intelligence**, **robotics**, **video games**, and even **movies**.
  (c) **CHEMISTRY** to calculate the rates of radioactive decay in, and even to predict birth and death rates.
  (d) **SHIP BUILDING** to determine both the curve of the hull of the ship (using differential calculus) and the area under the hull (using integral calculus).
  (e) **ECONOMICS** to predict supply, demand, and maximum potential profits.

PRE-CALCULUS	**CALCULUS**
Mathematics is more static	Mathematics is more dynamic
Object moving with constant speed	Accelerating object
Slope of straight line	Slope of a curve
Secant line to a curve	Tangent line to a curve
The curvature of a circle	The curvature of a curve
The area of geometric shapes	The area under a curve
Length of a line segment	Length of an arc
Surface area of a cylinder	Surface area of a solid of revolution
Mass of a solid of constant density	Mass of a solid of variable density
Volume of a rectangular solid	Volume of a region under a surface
Sum of a finite number of terms	Sum of an infinite number of terms

What then is a bridge that connects PRE-CALCULUS and CALCULUS in Mathematics?

PRE-CALCULUS ⇒ LIMIT PROCESS ⇒ CALCULUS

Finding Limits Graphically and Numerically

To sketch the graph of the function

$$f(x) = \frac{x^3 - 1}{x - 1}$$

for values other than $x \neq 1$, you can use standard curve-sketching techniques. At $x = 1$, however, it is not clear what to expect. To get an idea of the behavior of the graph of f near $x = 1$, you can use two sets of x-values - one set approaches 1 from the left and one set that approaches 1 from the right, as shown below.

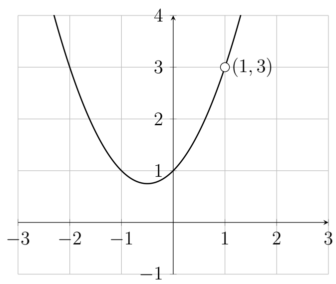
x	0.75	0.9	0.99	0.999	1	1.001	1.01	1.1	1.25
$f(x)$	2.313	2.710	2.9700	2.997	?	3.003	1	3.310	3.813

The graph of f is a parabola that has a gap at the point $(1, 3)$. Although x cannot equal 1, you can move arbitrarily close to 1, and as a result $f(x)$ moves arbitrarily close to 3.

The limit of f as x approaches 1 is 3. Using limit notation, you can write

$$\lim_{x \to 1} f(x) = 3$$

read as "**The limit of $f(x)$ as x approaches 1 is 3.** "



Definition 1. *If $f(x)$ becomes arbitrarily close to a single number L as x approaches c from either side, then the limit of f as x approaches c is L. This limit is written as*

$$\lim_{x \to c} f(x) = L.$$

Example 2. *Evaluate the function* $f(x) = \dfrac{x}{\sqrt{x+1}-1}$ *at several x-values near 0 and use the result to estimate the limit*

$$\lim_{x\to 0} \frac{x}{\sqrt{x+1}-1}.$$

The table below lists the values of $f(x)$ for several x-values near 0.

x	-0.01	-0.001	-0.0001	0	0.0001	0.001	0.01
$f(x)$	1.99499	1.99950	1.99995	?	2.00005	2.00050	2.00499

From the results shown in the table, you can estimate the limit to be 2; that is,

$$\lim_{x\to 0} \frac{x}{\sqrt{x+1}-1} = 2.$$

Example 3. *Find the limit of* $f(x) = \begin{cases} 1, & x \neq 2 \\ 0, & x = 2. \end{cases}$ *as x approaches 2.*

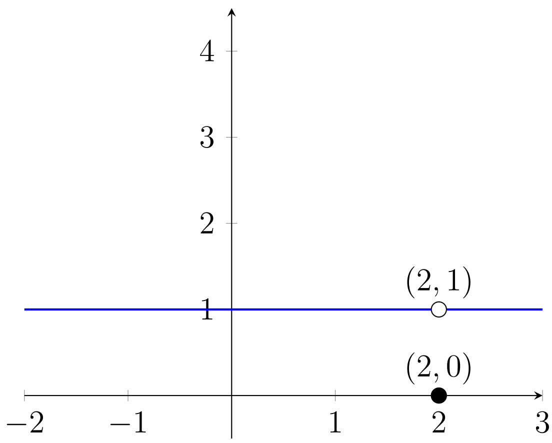
The table below lists the values of $f(x)$ for several x-values near 2.

x	1.9	1.99	1.999	2	2.0001	2.001	2.01
$f(x)$	1	1	1	?	1	1	1

From the results shown in the table, you can estimate the limit to be 1; that is,

$$\lim_{x\to 2} f(x) = 1.$$

Given the graph of f below.



From the graph of f we will get the same answer.

$$\lim_{x\to 2} f(x) = 1.$$

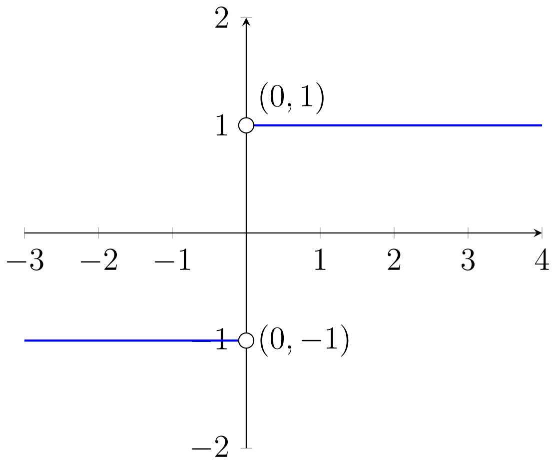
Example 4 (Different Right and Left Behavior)**.** *Consider the graph of the function*

$$f(x) = \frac{|x|}{x}.$$

It follows from absolute value definition that

$$f(x) = \frac{|x|}{x} = \begin{cases} 1, & x > 0 \\ -1, & x < 0. \end{cases}$$

The graph is given below.



It follows from the graph that

1. *$f(x)$ approaches to -1 when x approaches to 0 from the left.*

2. *$f(x)$ approaches to 1 when x approaches to 0 from the right.*

Therefore, there does not exist a single number L to which $f(x)$ approaches as x approaches 0 *from both sides. Hence $\lim_{x\to 0} f(x)$ does not exist.*

Common Types of Behavior Associated with Nonexistence of a Limit

1. $f(x)$ approaches a different number from the right side of c than it approaches from the left side.

2. $f(x)$ increases or decreases without bound as x approaches c. $[f(x) = \dfrac{1}{x^2}$ at $x = 0]$

3. $f(x)$ oscillates between two fixed values as x approaches c. $[f(x) = \cos(\dfrac{1}{x})$ at $x = 0]$

Numerical and Graphical approaches to limit produce an estimate of the limit. In the subsequent sections we will study analytic techniques for evaluating limits. Throughout the course, try to develop a habit of using this three-pronged approach to problem solving.

1. Numerical approach ... [Construct a table of values.]

2. Graphical approach ..... [Draw a graph by hand or using technology.]

3. Analytic approach ...... [Use algebra or calculus.]

Evaluating Limits Analytically

The limit of $f(x)$ as x approaches c does not depend on the value of f at c. It may happen, however, that the limit is precisely $f(c)$. In such cases, the limit can be evaluated by **direct substitution**. That is,

$$\lim_{x\to c} f(x) = f(c) \qquad \text{(Substitute c for x)}$$

Theorem 5 (Some Basic Limits)**.** *Let b and c be real numbers, and let n be a positive integer.*

$$
\begin{array}{lll}
(1) & \lim\limits_{x\to c} b = b & \text{Limit of a Constant function} \\
(2) & \lim\limits_{x\to c} x = c & \text{Limit of the identity function} \\
(3) & \lim\limits_{x\to c} x^n = c^n & \text{Limit of a power function}
\end{array}
$$

Example 6 (Evaluating Basic Limits)**.**

$$
\begin{array}{lll}
(a) \lim\limits_{x\to 3} 3 = 3 & (c) \lim\limits_{x\to 2} x^2 = 2^2 = 4 & (e) \lim\limits_{x\to -5} x^3 = (-5)^3 = -125 \\
(b) \lim\limits_{x\to -4} x = -4 & (d) \lim\limits_{x\to\ 0} \sqrt{5} = \sqrt{5} & (f) \lim\limits_{x\to -1} x^{10} = (-1)^{10} = 1
\end{array}
$$

Theorem 7 (Properties of Limits)**.** *Let b and c be real numbers, let n be a positive integer, and let $f(x)$ and $g(x)$ be functions with the limits*

$$\lim_{x\to c} f(x) = L \quad \text{and} \quad \lim_{x\to c} g(x) = K.$$

$$
\begin{array}{lll}
(1) & \text{Scalar multiple:} & \lim\limits_{x\to c}[bf(x)] = bL = b\lim\limits_{x\to c} f(x). \\
(2) & \text{Sum or difference:} & \lim\limits_{x\to c}[f(x) \pm g(x)] = L \pm K = \lim\limits_{x\to c} f(x) \pm \lim\limits_{x\to c} g(x). \\
(3) & \text{Product:} & \lim\limits_{x\to c}[f(x)g(x)] = LK = \lim\limits_{x\to c} f(x). \lim\limits_{x\to c} g(x). \\
(4) & \text{Quotient:} & \lim\limits_{x\to c}[\frac{f(x)}{g(x)}] = \frac{L}{K} = \frac{\lim_{x\to c} f(x)}{\lim_{x\to c} g(x)}, \qquad K \neq 0. \\
(5) & \text{Power:} & \lim\limits_{x\to c}[f(x)]^n = L^n = \left[\lim\limits_{x\to c} f(x)\right]^n.
\end{array}
$$

Example 8 (Limit of a Polynomial Function)**.** *Find the limit*

$$\lim_{x\to 2}(4x^2 + 3).$$

Solution.

$$
\begin{aligned}
\lim_{x\to 2}(4x^2 + 3) &= \lim_{x\to 2}(4x^2) + \lim_{x\to 2} 3 && \text{Sum Rule} \\
&= 4\lim_{x\to 2} x^2 + \lim_{x\to 2} 3 && \text{Scalar Multiple Rule} \\
&= 4(2^2) + 3 && \text{Power Rule} \\
&= 19 && \text{Simplify.}
\end{aligned}
$$

NB. Note that the limit (as x approaches 2) of the polynomial function $p(x) = 4x^2+3$ is simply the value of p at $x = 2$.

The direct substitution property is valid for all polynomial and rational functions with nonzero denominators.

Theorem 9. ***Limits of Polynomial and Rational Functions***

1. If p is a polynomial function and c is a real number, then

$$\lim_{x\to c} p(x) = p(c).$$

2. If r is a rational function given by

$$r(x) = \frac{p(x)}{q(x)}$$

and c is a real number such that $q(c) \neq 0$, then

$$\lim_{x\to c} r(x) = r(c) = \frac{p(c)}{q(c)}$$

Example 10 (Limit of a Rational Function)**.** *Find the limits*

$$(a) \quad \lim_{x\to 1} \frac{x^2+x+2}{x+1} \qquad\qquad (b) \quad \lim_{x\to 5} \frac{x-2}{2x-1}.$$

Solution.
(a) Here the numerator polynomial $p(x) = x^2 + x + 2$ and the denominator polynomial is $q(x) = x + 1$. Since $q(1) = 1 + 1 = 2 \neq 0$, we can apply the above Theorem to obtain:

$$\lim_{x\to 1} \frac{x^2+x+2}{x+1} = \frac{1^2+1+2}{1+1} = \frac{4}{2} = 2.$$

(b) Here the numerator polynomial $p(x) = x - 2$ and the denominator polynomial is $q(x) = 2x - 1$. Since $q(5) = 9 \neq 0$, we can apply the above Theorem to obtain:

$$\lim_{x\to 1} \frac{x-2}{2x-1} = \frac{3}{9} = \frac{1}{3}.$$

Theorem 11 (The Limit of a Function Involving a Radical)**.** *Let n be a positive integer. The limit below is valid for all c when n is odd, and is valid for $c > 0$ when n is even.*

$$\lim_{x\to c} \sqrt[n]{x} = \sqrt[n]{c}.$$

Theorem 12 (The Limit of a Composite Function)**.** *If f and g are functions such that $\lim_{x\to c} g(x) = L$ and $\lim_{x\to L} f(x) = f(L)$ then*

$$\lim_{x\to c}(f \circ g)(x) = \lim_{x\to c} f(g(x)) = f(\lim_{x\to c} g(x)) = f(L).$$

Remark 13 (Limits of Algebraic Functions)**.** *We have seen in Theorems 5, 6, and 7 the limits of the basic three **algebraic functions**: (1) polynomial functions, (2)rational functions, and (3) functions involving radicals.*

Example 14 (The Limit of a Composite Function)**.** *Find the limit*

$$(a) \quad \lim_{x\to 0} \sqrt{x^2+4} \qquad\qquad (b) \quad \lim_{x\to 3} \sqrt[3]{2x^2-10}$$

Solution.

(a) If $g(x) = x^2 + 4$ and $f(x) = \sqrt{x}$, then $(f \circ g)(x) = \sqrt{x^2+4}$. Since $\lim_{x\to 0}(x^2+4) = 4$ and $\lim_{x\to 4} \sqrt{x} = \sqrt{4} = 2$, we can apply Theorem 8 to obtain:

$$\lim_{x\to 0} \sqrt{x^2+4} = \lim_{x\to 0} f(g(x)) = f(\lim_{x\to 0} g(x)) = f(4) = 2.$$

(b) If $g(x) = 2x^2 - 10$ and $f(x) = \sqrt[3]{x}$, then $(f \circ g)(x) = \sqrt[3]{2x^2-10}$. Since $\lim_{x\to 3} g(x) = -8$ and $\lim_{x\to -8} f(x) = f(-8) = -2$, we can apply Theorem 8 to obtain:

$$\lim_{x\to 3} \sqrt[3]{2x^2-10} = -2.$$

Limits of Trigonometric Functions

We have seen that the limits of many algebraic functions (polynomial, rational, and radicals) can be evaluated by direct substitution. The six basic trigonometric functions also exhibit this desirable quality.

Theorem 15 (Limits of Trigonometric Functions)**.** *Let c be a real number in the domain of the given trigonometric function.*

1. $\lim_{x\to c} \sin x = \sin c$
2. $\lim_{x\to c} \cos x = \cos c$
3. $\lim_{x\to c} \tan x = \tan c$
4. $\lim_{x\to c} \cot x = \cot c$
5. $\lim_{x\to c} \sec x = \sec c.$
6. $\lim_{x\to c} \csc x = \csc c.$

Example 16 (Evaluating Limits of Trigonometric Functions)**.**

(a) $\lim_{x\to 0} \tan x = \tan 0 = 0$

(b) $\lim_{x\to \frac{\pi}{2}} \sin x = \sin\left(\frac{\pi}{2}\right) = 1$

(c) $\lim_{x\to \pi}(x\cos x) = (\lim_{x\to\pi} x)(\lim_{x\to\pi}\cos x) = \pi.\cos\pi = -\pi$

(d) $\lim_{x\to\ 0} \sin^2 x = \lim_{x\to\ 0}(\sin x)^2 = 0^2 = 0$

Strategies for Finding Limits

Theorem 17 (Functions That Agree at All but One Point)**.** *Let c be a real number, and let $f(x) = g(x)$ for all $x \neq c$ in an open interval containing c. If the limit of $g(x)$ as x approaches c exists, then the limit $f(x)$ of also exists and*

$$\lim_{x\to c} f(x) = \lim_{x\to c} g(x).$$

Example 18 (Finding the Limit of a Function)**.** *Find the limit*

$$\lim_{x\to 1}\frac{x^3-1}{x-1}.$$

Solution. *Note that for $x \neq 1$,*

$$\frac{x^3-1}{x-1} = \frac{(x-1)(x^2+x+1)}{x-1} = x^2+x+1.$$

Let $f(x) = \frac{x^3-1}{x-1}$ and $g(x) = x^2+x+1$. Then $f(x) = g(x)$ for all $x \neq 1$ in any open interval containing 1*. Therefore, by the above theorem,*

$$\lim_{x\to 1}\frac{x^3-1}{x-1} = \lim_{x\to 1} f(x) = \lim_{x\to 1} g(x) = \lim_{x\to 1} x^2+x+1 = 1+1+1 = 3.$$

Alternatively, we can find the limit as follows:

$$\begin{aligned}
\lim_{x\to 1}\frac{x^3-1}{x-1} &= \lim_{x\to 1}\frac{(x-1)(x^2+x+1)}{x-1} && \text{Factor} \\
&= \lim_{x\to 1}\frac{\cancel{(x-1)}(x^2+x+1)}{\cancel{x-1}} && \text{Divide out like factors} \\
&= \lim_{x\to 1}(x^2+x+1) && \text{Apply Theorem} \\
&= 1+1+1 && \text{Limit of polynomial function} \\
&= 3.
\end{aligned}$$

A Strategy for Finding Limits

1. Learn to recognize which limits can be evaluated by direct substitution.
2. When the limit $f(x)$ of x as approaches c cannot be evaluated by direct substitution, try to find a function that agrees with for all x other than $x = c$. [Choose g such that the limit of $g(x)$ can be evaluated by direct substitution.] Then apply Theorem 3 to conclude analytically that
$$\lim_{x\to c} f(x) = \lim_{x\to c} g(x) = g(c).$$
3. Use a graph or table to reinforce your conclusion.

Example 19 (Dividing Out Technique)**.** *Find the limit*
$$\lim_{x\to -3} \frac{x^2+x-6}{x+3}.$$

Solution. *Note that direct substitution produces the meaningless fractional form* $\frac{0}{0}$. *By factorization of the numerator we get*
$$x^2+x-6 = (x+3)(x-2).$$
Therefore,
$$\begin{aligned}
\lim_{x\to -3} \frac{x^2+x-6}{x+3} &= \lim_{x\to 1} \frac{(x+3)(x-2)}{x+3} && \text{Factor}\\
&= \lim_{x\to -3} \frac{\cancel{(x+3)}(x-2)}{\cancel{x+3}} && \text{Divide out like factors}\\
&= \lim_{x\to -3}(x-2) && \text{Apply Theorem}\\
&= -3-2 && \text{Limit of polynomial function}\\
&= -5.
\end{aligned}$$

Definition 20. *An expression of the form* $\frac{0}{0}$ *is called an* ***indeterminate form****.*

Example 21 (Rationalizing Technique)**.** *Find the limit*
$$\lim_{x\to 0} \frac{\sqrt{x+1}-1}{x}.$$

Solution. *By direct substitution, we obtain the indeterminate form* $\frac{0}{0}$.
$$\begin{aligned}
\lim_{x\to 0} \frac{\sqrt{x+1}-1}{x} &= \lim_{x\to 0} \frac{(\sqrt{x+1}-1)(\sqrt{x+1}+1)}{x(\sqrt{x+1}+1)} && \text{Rationalizing the numerator}\\
&= \lim_{x\to 0} \frac{(x+1)-1}{x(\sqrt{x+1}+1)}\\
&= \lim_{x\to 0} \frac{\cancel{x}}{\cancel{x}(\sqrt{x+1}+1)} && \text{Divide out like factors}\\
&= \lim_{x\to 0} \frac{1}{\sqrt{x+1}+1}\\
&= \frac{1}{\sqrt{1}+1} && \text{Limit of a function involving radicals}\\
&= \frac{1}{2}.
\end{aligned}$$

Reinforce the conclusion that $\lim_{x\to 0} \frac{\sqrt{x+1}-1}{x} = \frac{1}{2}$ *by constructing a table or a graph.*

Theorem 22 (The Squeeze/ or Sandwich/ or Pinching/ Theorem)**.** *If* $h(x) \le f(x) \le g(x)$ *for all* x *in an open interval containing* c*, except possibly at* c *itself, and if*

$$\lim_{x\to c} h(x) = L = \lim_{x\to c} g(x),$$

then $\lim_{x\to c} f(x)$ *exists and*

$$\lim_{x\to c} f(x) = L.$$

Theorem 23. ***Two Special Trigonometric Limits***

$$(1)\quad \lim_{x\to 0} \frac{\sin x}{x} = 1 \qquad\qquad (2)\quad \lim_{x\to 0} \frac{1-\cos x}{x} = 0$$

Example 24 (A Limit Involving a Trigonometric Function)**.** *Find the limit*

$$\lim_{x\to 0} \frac{\tan x}{x}.$$

Solution*. Direct substitution yields the indeterminate form* $\frac{0}{0}$*. To solve this problem, you can write* $\tan x$ *as* $\frac{\sin x}{\cos x}$ *and obtain*

$$\frac{\tan x}{x} = \frac{\sin x}{x\cos x} = \frac{\sin x}{x}\frac{1}{\cos x}.$$

Now, because $\lim_{x\to 0} \frac{\sin x}{x} = 1$ *and* $\lim_{x\to 0} \frac{1}{\cos x} = \frac{1}{\cos 0} = 1$*, we can obtain*

$$\begin{aligned}\lim_{x\to 0} \frac{\tan x}{x} &= \lim_{x\to 0} \frac{\sin x}{x\cos x}\\ &= (\lim_{x\to 0} \frac{\sin x}{x})(\lim_{x\to 0} \frac{1}{\cos x})\\ &= 1\end{aligned}$$

Example 25 (A Limit Involving a Trigonometric Function)**.** *Find the limit*

$$\lim_{x\to 0} \frac{\sin(4x)}{x}.$$

Solution*. Direct substitution yields the indeterminate form* $\frac{0}{0}$*. To solve this problem, you can rewrite the limit as*

$$\lim_{x\to 0} \frac{\sin(4x)}{x} = 4(\lim_{x\to 0} \frac{\sin(4x)}{4x}) \qquad \text{Multiply and divide by 4.}$$

Now, by letting $y = 4x$ *and observing that* y *approaches* 0 *if and only if* x *approaches* 0*, you can write*

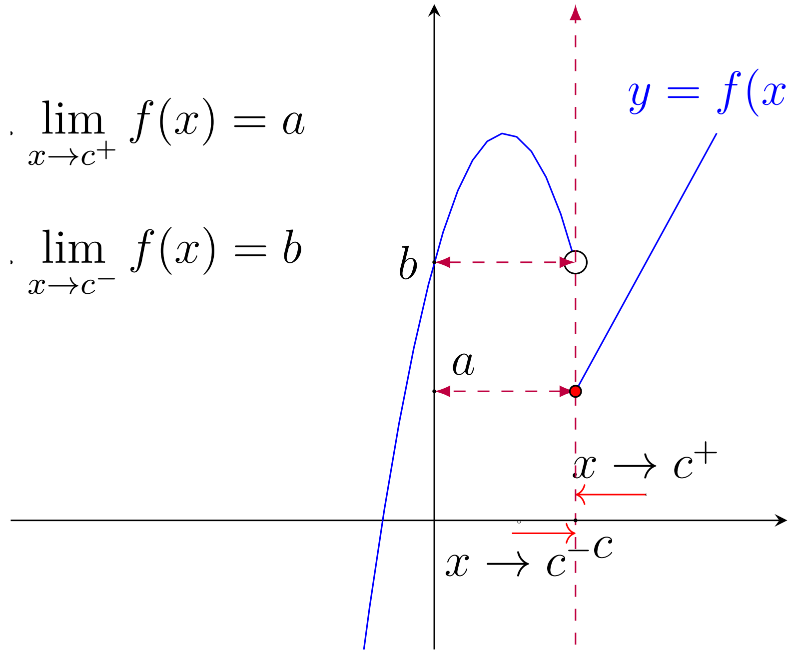
$$\begin{aligned}\lim_{x\to 0} \frac{\sin(4x)}{x} &= 4(\lim_{x\to 0} \frac{\sin 4x}{4x})\\ &= 4(\lim_{y\to 0} \frac{\sin y}{y}) \qquad \text{Let } y = 4x\\ &= 4\end{aligned}$$

One-Sided Limits

Definition 26 (One-Sided Limits)**.** *Suppose f is a function and c is a fixed real number.*

1. *A real number L is called **the left-hand limit** of f at c, written as $\lim_{x\to c^-} f(x) = L$, if and only if for all values of x sufficiently close to c from the left side of c (x approaches c from values less than c), the corresponding values of f approach to L.*

2. *A real number L is called **the right-hand limit** of f at c, written as $\lim_{x\to c^+} f(x) = L$, if and only if for all values of x sufficiently close to c from the right side of c (x approaches c from values greater than c), the corresponding values of f approach to L.*



One-sided limits can be used to investigate the behavior of step functions. One common type of step function is the greatest integer function $\lfloor . \rfloor$ defined as

$$\lfloor x \rfloor = \quad \text{greatest integer} \quad n \quad \text{such that} \quad n \leq x.$$

Example 27 (The Greatest Integer Function)**.** *Find the limit of the greatest integer function $f(x) = \lfloor x \rfloor$ as x approaches 0 from the left and from the right.*
Solution.

As shown in Figure on the right, the limit as x approaches 0 from the left is

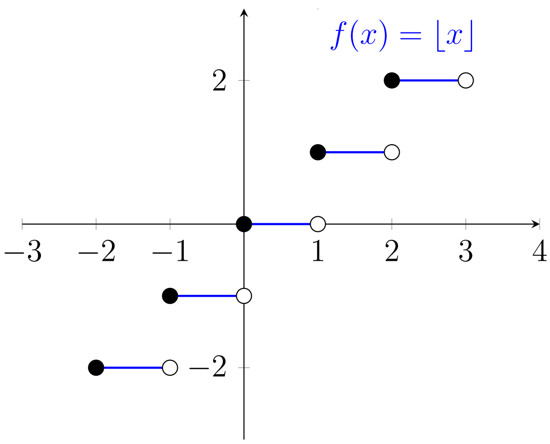
$$\lim_{x\to 0^-} f(x) = -1$$

and the limit as x approaches 0 from the right is

$$\lim_{x\to 0^+} f(x) = 0.$$

The greatest integer function has a discontinuity at zero because the left- and right-hand limits at zero are different. By similar reasoning, you can see that the greatest integer function has a discontinuity at any integer n.

Greatest Integer Function



Theorem 28 (The Existence of a Limit)**.** *Let f be a function, and let c and L be real numbers. The limit of $f(x)$ as x approaches c is L if and only if*

$$\lim_{x\to c^-} f(x) = L \quad \text{and} \quad \lim_{x\to c^+} f(x) = L.$$

INFINITE LIMITS

Infinite Limits from the left and from the right

Consider the function

$$f(x) = \frac{3}{x-2}.$$

Note that domain of f is $\mathbb{R} \setminus \{2\} = (-\infty, 2) \cup (2, \infty)$. Let us discuss the behavior of the function, numerically or graphically, as $x-$values approaches 2 from the left $(x - 2 < 0)$ and from the right$(x - 2 > 0.)$ Carefully examine the numerical tables and graph of the function f to get clear image of the nature of the values of f when x gets very close and close to 2 from the left or from the right.

1. The values of the function f as x approaches 2 from the left is given in the following table.

x	1	1.5	1.9	1.95	1.99	1.999	1.9999	1.99999	1.999999	1.9999999	2
$f(x)$	-3	-6	-30	-60	-300	-3,000	-30,000	-300,000	-3,000,000	-30,000,000	$\nexists$

From this table we observe that x goes from 1 t0 1.999999, (move close to 2 from the left in less than one unit) the value of f jumped from -3 to $-3,000,000$. This situation can be expressed as:

$f(x)$ **decreases without bound as** x **approaches** 2 **from the left.**

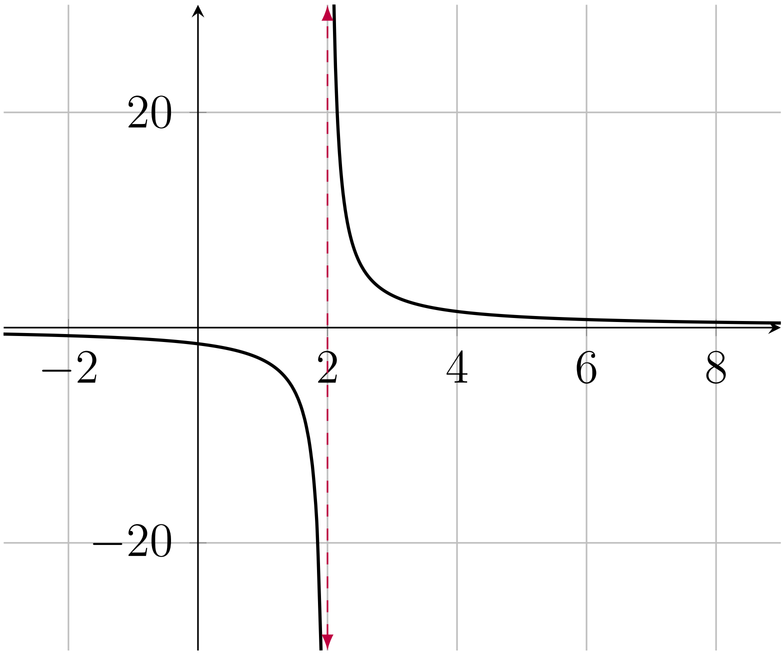
2. The values of the function f as x approaches 2 from the right is given in the following table.

x	2	2.00000003	2.0000001	2.000001	2.00003	2.0001	2.001	2.01	2.05	2.2
$f(x)$	$\nexists$	100,000,0000	30,000,000	3,000,000	100,000	30,000	3000	300	60	15

From this table we observe that x goes from 2.2 to 2.00000003, (move close to 2 from the left in less than one unit) the value of f jumped from 15 to $100,000,000$. This situation can be expressed as:

$f(x)$ **increases without bound as** x **approaches** 2 **from the right**.

In addition to the above numerical observation let us also investigate the nature of the graph of f when x approaches 2 from the left or from the right.



The graph justifies our numerical observations very well. We have the following mathematical notations for the behavior of the function f.

$\lim\limits_{x\to 2^-} f(x) = -\infty$ $f(x)$ decreases without bound as x approaches 2 from the left.

$\lim\limits_{x\to 2^+} f(x) = \infty$ $f(x)$ increases without bound as x approaches 2 from the right.

The symbol $-\infty$ and $+\infty$ refer to positive infinity and negative infinity, respectively. These symbols do not represent real numbers. They are convenient symbols used to describe unbounded conditions more concisely.

Definition 29 (Definition of Infinite Limits)**.** *Let f be a function that is defined at every real number x in some open interval containing c (except possibly at c itself). A limit in which $f(x)$ increases and decreases without bound as x approaches c is called an* ***infinite limit****. The statement*

1. $\lim\limits_{x\to c^-} f(x) = -\infty$ *means that $f(x)$ decreases without bound as x approaches c from the left.*

2. $\lim\limits_{x\to c^-} f(x) = +\infty$ *means that $f(x)$ increases without bound as x approaches c from the left.*

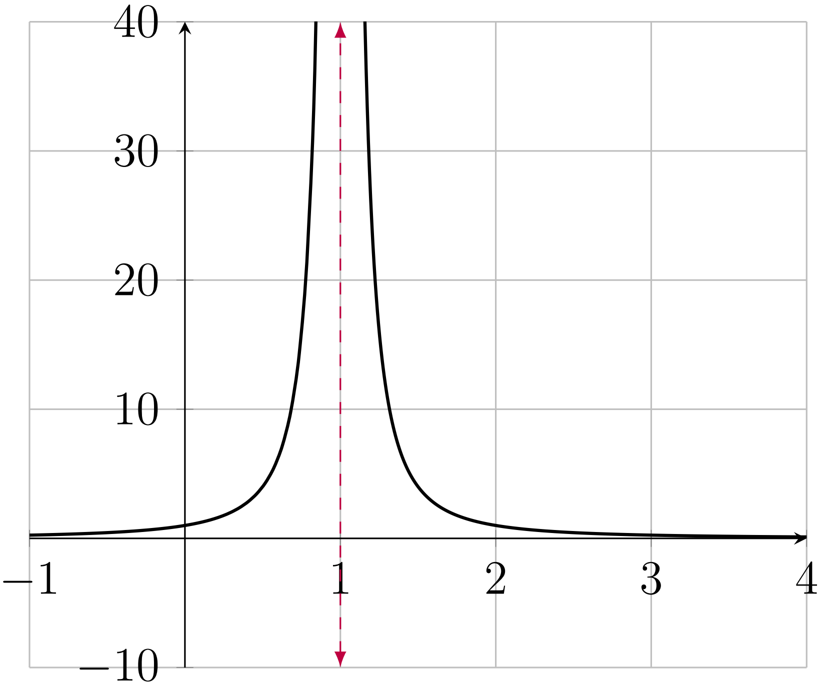
3. $\lim\limits_{x \to c^+} f(x) = -\infty$ *means that* $f(x)$ *decreases without bound as* x *approaches* c *from the right.*

4. $\lim\limits_{x \to c^+} f(x) = +\infty$ *means that* $f(x)$ *increases without bound as* x *approaches* c *from the right.*

5. $\lim\limits_{x \to c} f(x) = -\infty$ *means that* $f(x)$ ***decreases without bound as*** x ***approaches*** c.

6. $\lim\limits_{x \to c} f(x) = +\infty$ *means that* $f(x)$ ***increases without bound as*** x ***approaches*** c.

Example 30. *Determine the limit of the function in the Figure below as* x *approaches 1 from the left and from the right.*



It is simple observation that gives us:

$$(i)\ \lim_{x \to 1^-} \frac{1}{(x-1)^2} = +\infty \qquad (ii)\ \lim_{x \to 1^+} \frac{1}{(x-1)^2} = +\infty \qquad \lim_{x \to 1} \frac{1}{(x-1)^2} = +\infty$$

Example 31. *Use a graphing utility to graph each function. For each function, analytically find the single real number* c *that is not in the domain. Then graphically find the limit (if it exists) of* $f(x)$ *as* x *approaches* c *from the left and from the right.*

$$(a)\ \frac{3}{x-4} \qquad (b)\ \frac{1}{2-x} \qquad (c)\ \frac{2}{(x-3)^2} \qquad (d)\ \frac{-3}{(x+2)^2}$$

subsectionVertical Asymptotes

Definition 32. *If* $f(x)$ *approaches infinity (or negative infinity) as* x *approaches* c *from the right or the left, then the vertical line* $x = c$ *is called a* ***vertical asymptote*** *of the graph of* f.

Theorem 33 (Vertical Asymptotes)**.** *Let f and g be continuous on an open interval containing c. If $f(c) \neq 0, \quad g(c) = 0$, and there exists an open interval containing c such that $g(x) \neq 0$ for all $x \neq c$ in the interval, then the graph of the function*

$$h(x) = \frac{f(x)}{g(x)}$$

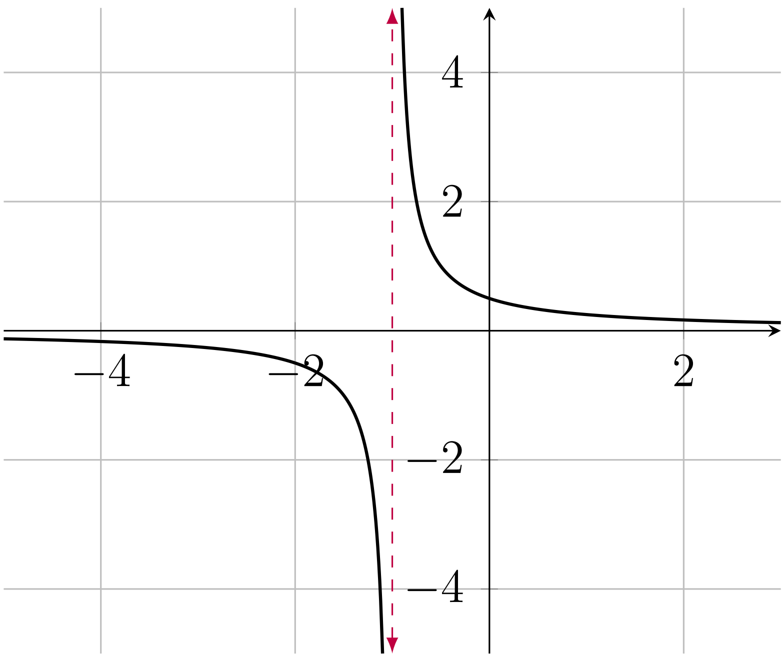
has a vertical asymptote at $x = c$.

Example 34. *Determine all vertical asymptotes of the graph of*

(a) $f(x) = \dfrac{1}{2(x+1)}$ (b) $f(x) = \dfrac{x^2+1}{x^2-1}$ (c) $f(x) = \cot x \dfrac{2}{(x-3)^2}$

Solution.

(a) *When $x = -1$ the denominator of $f(x) = \frac{1}{2(x+1)}$ is 0 and the numerator is not 0. So, by the Theorem, we can conclude that the line $x = -1$ is a vertical asymptote as shown in the figure below.*



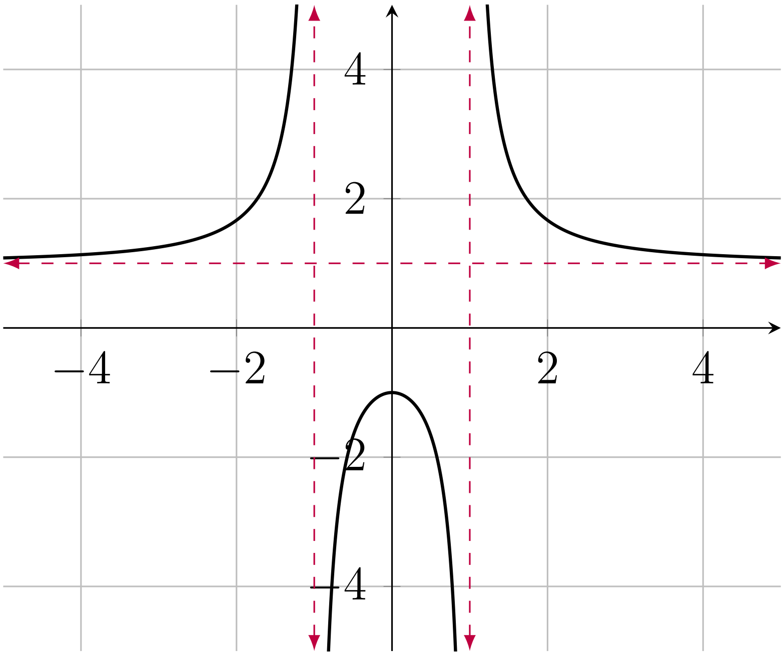
(b) *By factoring the denominator as*

$$f(x) = \frac{x^2+1}{x^2-1} = f(x) = \frac{x^2+1}{(x-1)(x-1)}$$

we see that the denominator is 0 at $x = 1$ and $x = -1$. Also, because the numerator is not 0 at these two points, we can apply the Theorem to conclude that the graph of f has two vertical asymptotes

$$x = -1 \text{ and } x = 1.$$

See the graph of f given below.

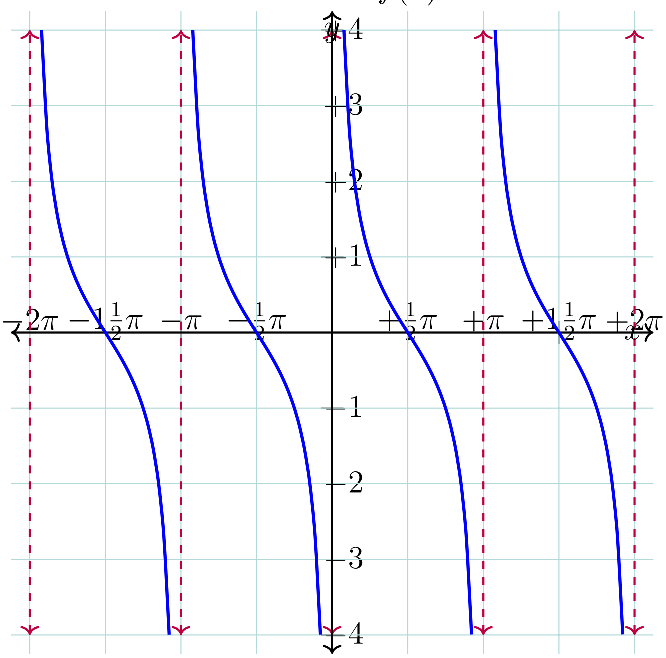


(c) By writing the cotangent function in the form

$$f(x) = \cot x = \frac{\cos x}{\sin x}$$

we can apply the theorem to conclude that vertical asymptotes occur at all values of x such that $\sin x = 0$. So, the graph of this function has infinitely many vertical asymptotes. These asymptotes occur at $x = n\pi$ where n is an integer. See the graph of $f(x) = \cot x$ given below to support our conclusion.

THE GRAPH OF $f(x) = \cot x$



LIMITS AT INFINITY

We consider the ”end behavior“ of a function on an infinite interval.

Definition 35. ***Intuitive Definition for Limit at Infinity***

1. *Let f be a function defined on some interval (a, ∞) and $L \in \mathbb{R}$. Then*

$$\lim_{x \to \infty} f(x) = L,$$

read as "the limit of $f(x)$ as x approaches ∞ is L," means that the values of $f(x)$ becomes arbitrarily close to L as x takes on large positive numbers.

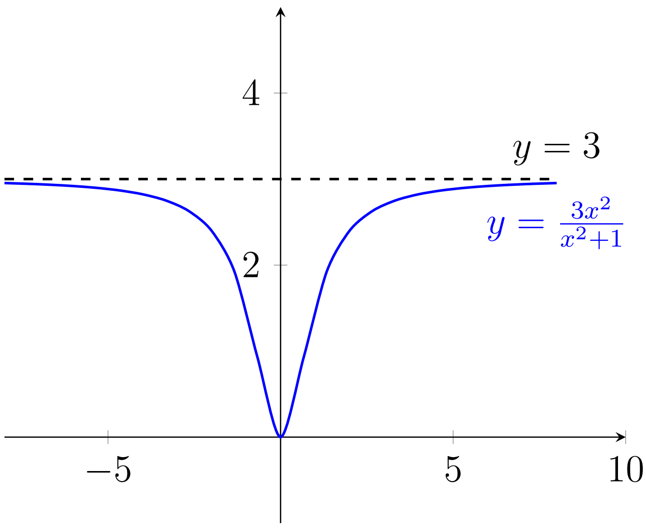
2. *Let f be a function defined on some interval $(-\infty, a)$ and $L \in \mathbb{R}$. Then*

$$\lim_{x \to -\infty} f(x) = L,$$

read as "the limit of $f(x)$ as x approaches $-\infty$ is L," means that the values of $f(x)$ becomes arbitrarily close to L as x takes on large negative numbers.

Remark 36. *The basic rules of limit, the substitution theorem and squeezing theorem are valid for limits at infinity when $x \to c$ is replaced by $x \to \infty$ or $x \to -\infty$.*

Example 37. *Find limits at infinity of the function $f(x) = \frac{3x^2}{x^2+1}$.* ***Solution.*** *Consider the graph of the function given below.*



Graphically, we can see that the values of $f(x)$ appear to approach 3 as x increases without bound or decreases without bound.

Numerically, we can construct the table as shown below.

x	$-\infty \leftarrow$	-1000	-100	-10	-1	0	1	10	100	1000	$\to \infty$
$f(x)$	$3 \leftarrow$	2.999997	2.9997	2.9703	1.5	0	1.5	2.9703	2.9997	2.999997	$\to 3$

The table suggests that the value of $f(x)$ approaches 3 as x increases without bound ($x \to \infty$). Similarly, $f(x)$ approaches 3 as x decreases without bound ($x \to -\infty$).

Therefore, limits at infinity are given as

$$\lim_{x \to -\infty} f(x) = 3 \qquad \text{Limit at negative infinity}$$

and

$$\lim_{x \to \infty} f(x) = 3 \qquad \text{Limit at positive infinity}$$

Theorem 38 (Limits at Infinity)**.** *If r is a positive rational number and c is any real number, then*

$$\lim_{x\to\infty} \frac{c}{x^r} = 0.$$

Furthermore, if x^r is defined when $x < 0$, then

$$\lim_{x\to-\infty} \frac{c}{x^r} = 0.$$

Example 39 (Finding a Limit at Infinity)**.** *Find the limits:* $\lim_{x\to\infty}\left(2+\frac{3}{x^2}\right)$ *and* $\lim_{x\to-\infty}\left(2+\frac{3}{x^2}\right)$.
Solution*. Let* $f(x) = 2 + \frac{3}{x^2}$.

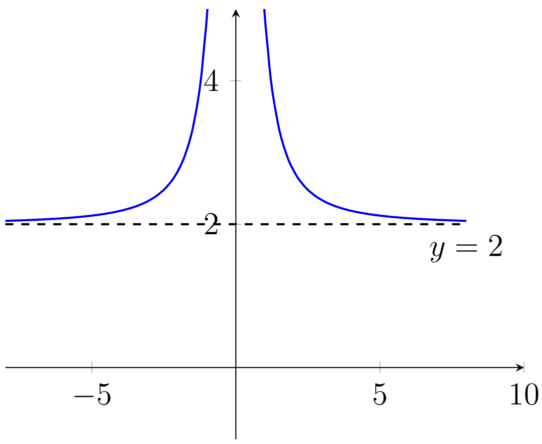
I. Numerical approach

x	$-\infty \leftarrow$	-1000	-100	-10	-1	0	1	10	100	1000	$\to\infty$
$f(x)$	$2 \leftarrow$	2.000003	2.0003	2.03	5	$\nexists$	5	2.03	2.0003	2.000003	$\to 2$

From the table we conclude that

$$\lim_{x\to-\infty}\left(2+\frac{3}{x^2}\right) = 2 \quad \text{and} \quad \lim_{x\to\infty}\left(2+\frac{3}{x^2}\right) = 2$$

II. Graphical approach: Consider the graph of $f(x) = 2 + \frac{3}{x^2}$.



The graph suggests that the value of $f(x)$ approaches 2 as x increases without bound ($x \to \infty$). Similarly, $f(x)$ approaches 2 as x decreases without bound ($x \to -\infty$).

Definition 40 (Horizontal Asymptote)**.** *The horizontal line $y = L$ is called a* ***horizontal asymptote*** *of the graph of f, when*

$$\lim_{x\to-\infty} f(x) = L \quad \text{or} \quad \lim_{x\to\infty} f(x) = L.$$

Example 41. *Evaluate the following limits at infinity.*

1. $\lim_{x\to\infty}\left(\frac{1}{x^3} - \frac{2}{x^2} + 3\right) = \lim_{x\to\infty}\frac{1}{x^3} - \lim_{x\to\infty}\frac{2}{x^2} + \lim_{x\to\infty} 3 = 0 - 0 + 3 = 3$

2. $\lim_{x\to\infty}\left(\frac{2x-3}{x+2}\right) = \lim_{x\to\infty}\left(\frac{2-\frac{3}{x}}{1+\frac{2}{x}}\right) = \frac{2+0}{1+0} = 2.$

Example 42. *Find horizontal asymptotes of the graph of the following functions.*

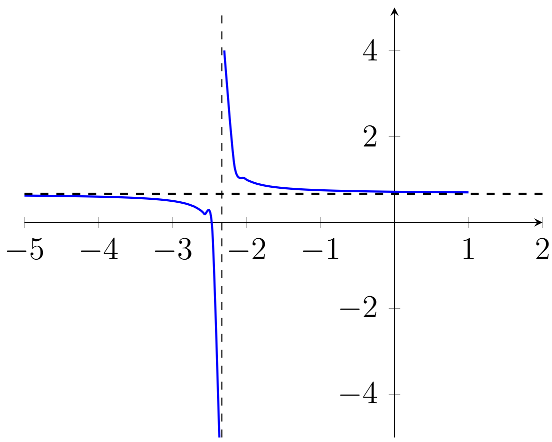
$$(a) \quad f(x) = \frac{2x+5}{3x+7} \qquad\qquad (b) \quad f(x) = \frac{x+5}{2x^2-1}$$

Solution.

(a) We note that f is defined on the interval $(0, \infty)$. Then

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty}\left(\frac{x+5}{3x+7}\right) = \lim_{x\to\infty}\left(\frac{2+\frac{5}{x}}{3+\frac{7}{x}}\right) = \frac{2+0}{3+0} = \frac{2}{3}.$$

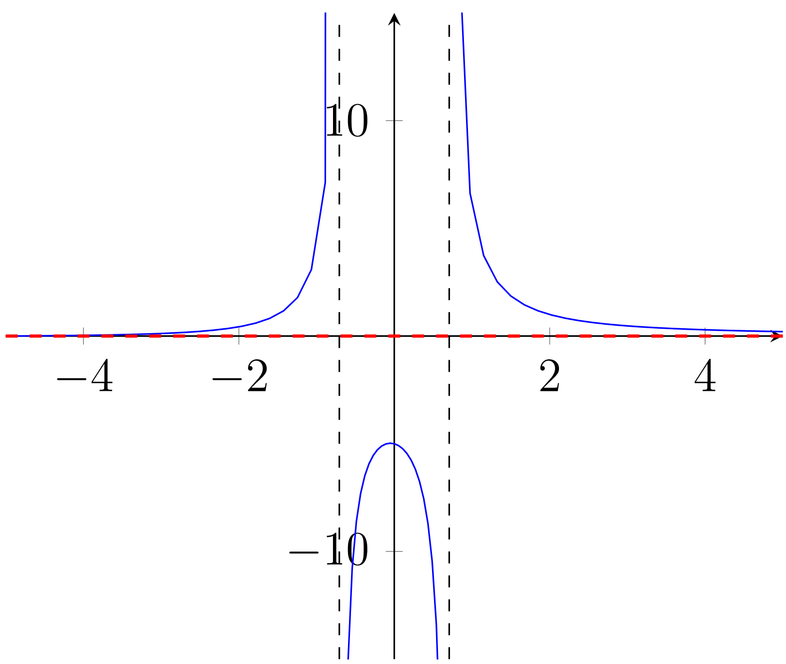
We also note that f is defined on $(-\infty, -3)$ and $\lim_{x\to\infty} f(x) = \frac{2}{3}$. Therefore, $y = \frac{2}{3}$ is the horizontal asymptote of the graph of the function. See the graph of f given below.



(b) We note that f is defined on the interval $(1, \infty)$. Then

$$\lim_{x\to\infty} f(x) = \lim_{x\to\infty}\left(\frac{x+5}{2x^2-1}\right) = \lim_{x\to\infty}\left(\frac{\frac{1}{x}+\frac{5}{x^2}}{2-\frac{1}{x^2}}\right) = \frac{0+0}{2-0} = 0.$$

We also note that f is defined on $(-\infty, -3)$ and $\lim_{x\to\infty} f(x) = 0$. Therefore, $y = 0$ is the horizontal asymptote of the graph of f. See the graph of f given below.



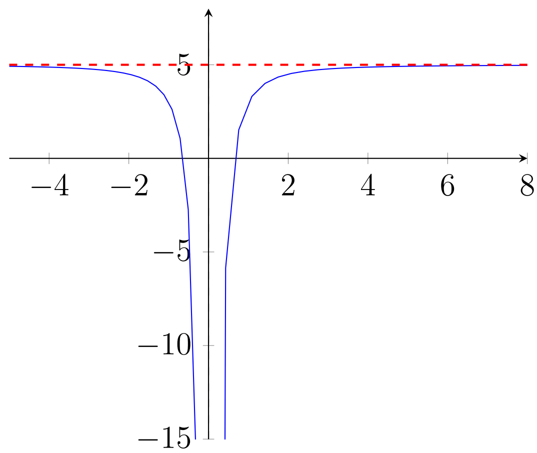
Example 43. *Find* $\lim_{x\to\infty}\left(5-\frac{2}{x^2}\right)$.
Solution. *Using the Difference rule for limit, you can write*

$$\lim_{x\to\infty}\left(5-\frac{2}{x^2}\right)=\lim_{x\to\infty}5-\lim_{x\to\infty}\frac{2}{x^2}=5-0=5$$

So, the line $y=5$ is a horizontal asymptote to the right. By finding the limit

$$\lim_{x\to-\infty}\left(5-\frac{2}{x^2}\right)=\lim_{x\to-\infty}5-\lim_{x\to-\infty}\frac{2}{x^2}=5-0=5$$

you can see that $y=5$ is also a horizontal asymptote to the left. The graph of the function is shown below.



Example 44 (A Comparison of Three Rational Functions)**.** *Find each limit.*

$$(a)\quad \lim_{x\to-\infty}\frac{2x+5}{3x^2+1}\qquad (b)\quad \lim_{x\to-\infty}\frac{2x+5}{3x^2+1}\qquad (c)\quad \lim_{x\to-\infty}\frac{2x^3+5}{3x^2+1}$$

Solution. *In each case, attempting to evaluate the limit produces the indeterminate form* $\frac{\infty}{\infty}$.

(a) Divide both the numerator and the denominator by x^2

$$\lim_{x\to-\infty}\frac{2x+5}{3x^2+1}=\lim_{x\to-\infty}\left(\frac{\frac{2}{x}+\frac{5}{x^2}}{3+\frac{1}{x^2}}\right)=\frac{0+0}{3+0}=\frac{0}{3}=0$$

(b) Divide both the numerator and the denominator by x^2

$$\lim_{x\to-\infty}\frac{2x^2+5}{3x^2+1}=\lim_{x\to-\infty}\left(\frac{2+\frac{5}{x^2}}{3+\frac{1}{x^2}}\right)=\frac{2+0}{3+0}=\frac{2}{3}$$

(c) Divide both the numerator and the denominator by x^2

$$\lim_{x\to-\infty}\frac{2x^3+5}{3x^2+1}=\lim_{x\to-\infty}\left[\frac{2x+\frac{5}{x^2}}{3+\frac{1}{x^2}}\right]=\frac{\infty}{3}$$

We can conclude that the limit does not exist because the numerator increases without bound while the denominator approaches 3.

Guidelines for finding limits at $\pm\infty$ of rational functions

1. If the degree of the numerator is less than the degree of the denominator, then the limit of the rational function is 0.

2. If the degree of the numerator is equal to the degree of the denominator, then the limit of the rational function is the ratio of the leading coefficients.

3. If the degree of the numerator is greater than the degree of the denominator, then the limit of the rational function does not exist.

Infinite Limits at Infinity

Many functions do not approach a finite limit as x increases (or decreases) without bound.

Definition 45. ***Infinite Limits at Infinity***

1. Let f be a function defined on the interval (a, ∞).

(a) The statement $\lim_{x\to\infty} f(x) = +\infty$ means that $f(x)$ increases without bound $(f(x) \to \infty)$, when x increases without bound $(x \to \infty)$.

(b) The statement $\lim_{x\to\infty} f(x) = -\infty$ means that $f(x)$ decreases without bound $(f(x) \to -\infty)$, when x increases without bound $(x \to \infty)$.

2. Let f be a function defined on the interval $(-\infty, a)$.

(a) The statement $\lim_{x\to-\infty} f(x) = +\infty$ means that $f(x)$ increases without bound $(f(x) \to \infty)$, when x decreases without bound $(x \to -\infty)$.

(b) The statement $\lim_{x\to-\infty} f(x) = -\infty$ means that $f(x)$ decreases without bound $(f(x) \to -\infty)$, when x decreases without bound $(x \to -\infty)$.

Example 46. *Using numerical approach find the following limit.*

$$(a) \quad \lim_{x\to\infty} x^3 \qquad\qquad (b) \quad \lim_{x\to-\infty} x^3$$

Solution. *Numerically, we can construct the table as shown below.*

x	$-\infty \leftarrow$	*-1000*	*-100*	*-10*	*-1*	*0*	*1*	*10*	*100*	*1000*	$\to \infty$
$f(x) = x^3$	$-\infty \leftarrow$	-10^9	-10^6	*-1000*	*-1*	*0*	*1*	*1000*	10^6	10^9	$\to \infty$

The table suggests that the value of

i. $f(x)$ increases without bound $(f(x) \to \infty)$ as x increases without bound $(x \to \infty)$.

ii. $f(x)$ decreases without bound$(f(x) \to -\infty)$ as x decreases without bound$(x \to -\infty)$.

Therefore, infinite limits at infinity are given as

$$\lim_{x\to-\infty} x^3 = +\infty \quad \text{and} \quad \lim_{x\to-\infty} f(x) = -\infty$$

Example 47. *Find each limit.*

$$(a) \quad \lim_{x\to\infty} \frac{2x^2-4x}{x+1} \qquad\qquad (b) \quad \lim_{x\to-\infty} \frac{2x^2-4x}{x+1}$$

Solution. *One way to evaluate each of these limits is to use long division to rewrite the improper rational function as the sum of a polynomial and a rational function.*

(a) $\lim_{x\to\infty} \frac{2x^2-4x}{x+1} = \lim_{x\to\infty} \left(2x-6+\frac{6}{x+1}\right) = +\infty.$

(b) $\lim_{x\to-\infty} \frac{2x^2-4x}{x+1} = \lim_{x\to-\infty} \left(2x-6+\frac{6}{x+1}\right) = -\infty.$

The statements above can be interpreted as saying that as x approaches $\pm\infty$, the function $f(x) = \frac{2x^2-4x}{x+1}$ behaves like the function $g(x) = 2x-6$. The line $y = 2x-6$ is called a ***slant asymptote*** *of the graph of f. [See the graph below.]*